

SOP 203: Incidental Findings

1 Summary

On rare occasions MRI research personnel may identify neurologic idiosyncracies, termed an ‘incidental finding’, in data collected at the Notre Dame Human Neuroimaging Center (ND-HNC). This document establishes procedures for responding to such incidental findings.

2 Scope

These policies apply to all investigators, research assistants, and staff who collect, review, or use MRI data collected at the ND-HNC.

3 Definitions

- **DICOM:** Digital Imaging and Communications in Medicine, a DICOM file consists of a header and image intensity data and is the foundational format for MRI data.
- **IRB:** Institutional Review Board, an independent committee that reviews and approves research involving human subjects to protect their rights, safety, and welfare.
- **PI:** Principal Investigator, the lead researcher of the project that takes intellectual, fiscal, and administrative responsibility.
- **Operator:** MRI Technologist or Level 4 researcher, responsible for conducting research operations and maintaining equipment. The Lead during emergency responses.

4 Policies & Procedures

The ND-HNC operates as a research facility focused on executing experimental MRI protocols for the purpose of scientific investigation. To this end, we provide personnel, infrastructure, and expertise that are appropriate to manage such research data. The ND-HNC is not a clinical facility and we are not organized to deploy clinical MRI protocols for the purpose of medical diagnosis. Given the volume of scans processed at the facility, however, research personnel will occasionally encounter idiosyncratic or anomalous anatomy.

Encountering such idiosyncracies introduces a number of responsibilities which the researcher must navigate to fulfill ethical and legal obligations. Namely, research personnel:

- do not normally use diagnostic MRI protocols nor hold the necessary qualifications to make medical diagnoses,

- must not cause harm to participants by *commission* (e.g. false alarm leading to undue stress or unnecessary medical expenses) or *omission* (e.g. withholding information which would have been medically relevant),
- are required to protect participant anonymity, and
- must adhere to IRB regulations of data usage and governance.

To help research personnel navigate these obligations, we provide an escalation protocol to be followed whenever incidental findings are detected. A description of these procedures should be included in any IRB application involving use of the ND-HNC MRI scanner. In short, the ND-HNC helps the Principal Investigator (PI) seek a recommendation from a qualified clinician.

4.1 Initial Review

Research personnel may notice something unusual during the normal acquisition, review, or use of the research data. When an unexpected observation occurs:

- Research personnel should **NOT** share their concerns with the individual who contributed MRI data.
- The individual is **NOT** contacted for follow-up scans nor are they informed that an incidental finding has been detected and the escalation process started.
- The PI should be contacted about the incidental finding. It is the responsibility of the PI to coordinate with ND-HNC staff for External Review and with the participant for any potential Follow-Up.

4.2 External Review

The ND-HNC contracts with qualified clinicians [**TBD**] to review incidental findings. After a PI contacts ND-HNC staff to initiate an External Review:

- Participant structural DICOM files are copied to [**TBD**], an internal, short-term data storage location.
- The clinician reviews the DICOM files either on-site or remotely via the University's secure network.
- The clinician sends their review and recommendations to the director (Aron Barbey: abarbey@nd.edu) and MRI technologist (Jeff Fox: jfox22@nd.edu).

4.3 Follow-Up

The MRI technologist coordinates with the PI once a clinical review and recommendation is received; the director oversees this coordination. In cases where clinical follow-up is recommended, the role of PI is to communicate the clinician's recommendation.

If a clinical follow-up is **NOT** recommended:

- The DICOM files are removed from [TBD].
- No further action is necessary.

Whereas, if a clinical follow-up **IS** recommended:

- The PI schedules a meeting with the participant to share the clinician's report, referencing the following script:
 1. "Thank you for meeting with me today."
 2. "You recently participated in research at the Notre Dame Human Neuroimaging Center and part of the data you contributed were MRI scans."
 3. "While the data were collected for research purposes, something unexpected showed up in your scans."
 4. "Per our Standard Operating Procedures, your scans were referred to a radiologist for interpretation."
 5. Provide the radiologists review and recommendation.
 6. Conclude with "I am not a physician, so it is not appropriate for me to answer any questions about the radiologist's report or provide any advice."
- The DICOM files are stored at [TBD] an internal, long-term data storage location, along with the review notes, in case they are requested in the future.

5 Figures

None.